

- 23** When a frictionless and well-insulated bicycle pump is used to pump up a tyre, the air in the tyre becomes hotter than the surrounding air.

Which of the following statements best explains this observation?

- A** After compression, the air molecules collide more frequently with the walls of the tyre.
- B** Work is done on the air and the internal energy remains unchanged.
- C** Work is done on the air and since little thermal energy escapes, the internal energy increases.
- D** The internal energy increases because thermal energy is supplied and work is done on the air.

23	C	Work is done on the air via pumping action. However, there is no heat supplied to the air. Hence by the 1 st law of thermodynamics, the internal energy increases.
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24 A straight horizontal current-carrying wire lies at right angles to a horizontal magnetic field